

**GIRRAWEE HIGH SCHOOL**  
**MATHEMATICS EXTENSION 2 TASK 1**  
**DECEMBER 2009**

Instructions:

- \* Start each new question on a new piece of paper.
  - \* Use only one side of the paper.
  - \* Show all necessary working.
  - \* Marks may be deducted for careless or badly arranged work.
  - \* Approved Scientific calculators may be used.
- Time allowed 90 minutes

**Question 1. 8 Marks**

**MARKS**

Given  $z = 2 - 3i$  and  $w = -1 + 4i$  find in the form  $a + ib$

- |     |                 |   |
|-----|-----------------|---|
| (a) | $z + w$         | 1 |
| (b) | $zw$            | 2 |
| (c) | $\overline{zw}$ | 1 |
| (d) | $z^2$           | 2 |
| (e) | $\frac{1}{w}$   | 2 |

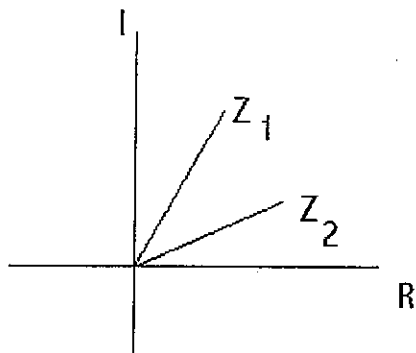
**Question 2. 16 Marks**

Given that  $z = \sqrt{3} - i$  and  $w = -1 + i$

- |     |                                                                                    |   |
|-----|------------------------------------------------------------------------------------|---|
| (a) | (i) Express both $z$ and $w$ in modulus argument form                              | 4 |
|     | (ii) Express $\frac{z^3}{w^4}$ in the form $a + ib$                                | 3 |
| (b) | (i) Find all real numbers $x$ and $y$ such that $\sqrt{8 + 6i} = x + iy$           | 3 |
|     | (ii) Hence or otherwise solve for $z$ : $z^3 + (1 + 3i)z^2 - 4z = 0$               | 3 |
| (c) | Find the fourth roots of $2\sqrt{3} + 2i$ you can leave the answer in mod/arg form | 3 |

### Question 3. 6 Marks

Copy the diagram below into your answer book,  $z_1$  and  $z_2$  represent unit vectors



(a) Draw the vectors representing

- |                        |   |
|------------------------|---|
| (i) $z_1 + z_2$        | 1 |
| (ii) $\frac{1}{z_1}$   | 1 |
| (iii) $\overline{z_2}$ | 1 |
| (iv) $-z_2$            | 1 |
| (v) $z_1 - z_2$        | 1 |
| (vi) $(z_1)^2$         | 1 |

### Question 4 20 Marks

(a) Sketch on separate Argand diagrams

- |                                      |   |
|--------------------------------------|---|
| (i) $ z - (2 + i)  \leq 2$           | 2 |
| (ii) $ z - (2 + i)  \leq  z - 2 $    | 2 |
| (iii) $\arg(z - 2i) = \frac{\pi}{3}$ | 2 |
| (iv) $ z^2 - (\bar{z})^2  \geq 4$    | 3 |

(b) Express the following loci in Cartesian form

- |                                                                                      |   |
|--------------------------------------------------------------------------------------|---|
| (i) $ z - (2 + i)  = 3$                                                              | 2 |
| (ii) $ z - 2  =  z + 2i $                                                            | 2 |
| (iii) $\arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{4}$ (Hint all details required) | 4 |

(c) The point  $P$  on the Argand plane represents the complex number  $z$  where

- |                                                                                          |   |
|------------------------------------------------------------------------------------------|---|
| $z$ satisfies $\frac{1}{z} + \frac{1}{\bar{z}} = 1$ Find the locus of $P$ as $z$ varies. | 3 |
|------------------------------------------------------------------------------------------|---|

**Question 5. 11 Marks**

Let  $z = \cos \theta + i \sin \theta$  by expanding  $z^4$  using De Moivre and binomial find expressions for

- (a)  $\cos 4\theta$  2
- (b)  $\sin 4\theta$  2
- (c)  $\tan 4\theta$  in terms of  $\tan \theta$  2
- (d) Prove that  $z^n + \frac{1}{z^n} = 2 \cos n\theta$  1
- (e)  $z^n - \frac{1}{z^n} = 2i \sin n\theta$  1
- (f) Show that  $\cos^6 \theta = \frac{1}{32}(\cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10)$  3

**Question 6. 14 Marks**

- (a) (i) Prove that  $\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} = \sin \theta + i \cos \theta$  3
  - (ii) Hence or otherwise deduce that  $(1 + \sin \frac{\pi}{5} + i \cos \frac{\pi}{5})^5 + i(1 + \sin \frac{\pi}{5} - i \cos \frac{\pi}{5})^5 = 0$  3
- (b) Let  $w = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$ 
  - (i) Show that  $w^k$  is a solution to  $z^5 - 1 = 0$  where  $k$  is an integer 2
  - (ii) Explain why  $1 + w + w^2 + w^3 + w^4 = 0$  2
  - (iii) Hence or otherwise find the value of  $(w-1)(1+2w+3w^2+4w^3+5w^4)$  2
  - (iv) Given that  $S = \frac{w}{1-w^2} + \frac{w^2}{1-w^4} + \frac{w^3}{1-w} + \frac{w^4}{1-w^3}$  Show that  $S = 0$  2

**Question 7. 14 Marks**

- (a) If  $n$  is a positive integer prove that  $\left( \frac{1+i \tan \theta}{1-i \tan \theta} \right)^n = \frac{1+i \tan n\theta}{1-i \tan n\theta}$  3
- (b) (i) Solve the equation  $z^6 - 1 = 0$  3
  - (ii) Hence factorise  $z^6 - 1$  into linear and quadratic factors with real coefficients 4
  - (iii) Sketch the roots of  $z^6 - 1 = 0$  onto a Argand diagram. 2
  - (iv) Find the area of the regular polygon formed by connecting the roots. 2

**Question 8. 9 Marks**

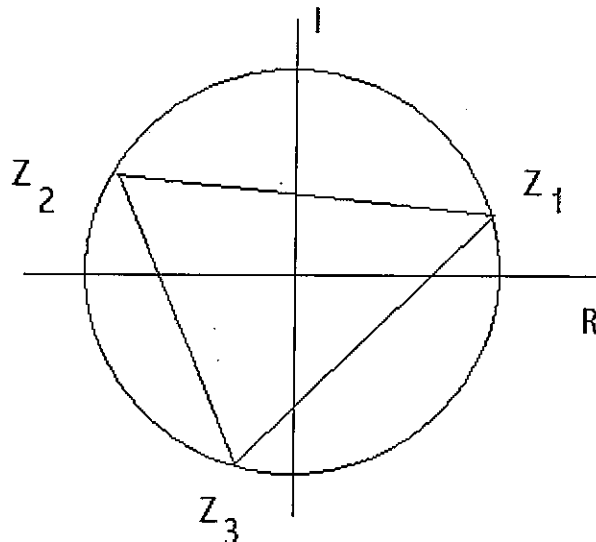
- (a) Determine the greatest and least values of  $\arg(z)$ , where  $|z - 4i| = 2$  3
- (b) If  $z = \cos \theta + i \sin \theta$  prove that  $\frac{2}{1+z} = 1 - i \tan \frac{\theta}{2}$  3
- (c) OABC is a square  $\overrightarrow{OA}$  is represented by the vector  $3+i$ . Find the vectors representing  $\overrightarrow{OB}$  and  $\overrightarrow{OC}$  3

**Question 9. 9 Marks**

Using the fact that  $|z|^2 = z\bar{z}$

- (a) Show that  $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2$  2
- (b) (i) Express  $1 + \cos \theta + i \sin \theta$  in modulus argument form. 3
- (ii) Find  $(1 + \cos \theta + i \sin \theta)^2$  in modulus argument form. 1
- (c) (i) Express  $\sin \theta + i \cos \theta$  in modulus argument form. 2
- (ii) Hence find  $\sqrt{\sin \theta + i \cos \theta}$  in modulus argument form. 1

**Question 10. 7 Marks**



$Z_1$ ,  $Z_2$  and  $Z_3$  are three points on the unit circle forming an equilateral triangle.

- (a) (i) If  $Z_1 = \cos \theta + i \sin \theta$  express  $Z_2$  and  $Z_3$  in terms of  $\theta$  2
- (ii) Prove that  $Z_1 + Z_2 + Z_3 = 0$  2
- (iii) Prove that  $Z_1^2 + Z_2^2 + Z_3^2 = Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1$  3

Q1  $z = 2 - 3i$   $w = -1 + 4i$

(a)  $z + w = 1 + i$  (1)

(b)  $zw = (2 - 3i)(-1 + 4i)$   
 $= -2 + 12 + 3i + 8i$   
 $= 10 + 11i$  (2)

(c)  $\overline{zw} = 10 - 11i$  (1)

(d)  $z^2 = 4 - 9 - 6i$   
 $= -5 - 6i$  (2)

(e)  $\frac{1}{w} = \frac{\overline{w}}{w\overline{w}} = \frac{-1 - 4i}{1 + 16}$   
 $= \frac{-1}{17} - \frac{4i}{17}$  (2)

Question 2

$z = \sqrt{3} - i$   $w = -1 + i$

(a)(i)  $|z| = \sqrt{(\sqrt{3})^2 + (-1)^2} = 2$   
 $\arg z = \tan^{-1} \frac{-1}{\sqrt{3}} = -\frac{\pi}{6}$

$|w| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$

(arg  $w = \tan^{-1} \frac{1}{-1} = \frac{3\pi}{4}$ )

$z = 2 \cos(-\pi/6)$   $w = \sqrt{2} \cos 3\pi/4$  (4)

(ii)  $\frac{z^3}{w^4} = \frac{8 \cos -\pi/2}{4 \cos 3\pi}$   $\cos 3\pi \equiv \cos \pi$   
 $= 2 \cos -3\pi/2$   
 $= 2 \cos \pi/2$   
 $= 2i$  (3)

(b)  $\sqrt{8 + 6i} = x + iy$

$\therefore 8 + 6i = x^2 - y^2 + 2xyi$   
 $8 = x^2 - y^2$  (A)  
 $6 = 2xy$

$5 = xy$  (B)

$\therefore y = \frac{5}{x}$

$\therefore 8 = x^2 - \frac{9}{x^2}$

$8x^2 = x^4 - 9$

$0 = x^4 - 8x^2 - 9$

$= (x^2 - 9)(x^2 + 1)$

$\therefore x = \pm 3$

$\therefore y = \pm 1$

$\sqrt{8 + 6i} = \pm (3 + i)$  (3)

(ii)  $z^3 + (1 + 3i)z^2 - 4z = 0$

$z(z^2 - (1 + 3i)z - 4) = 0$

$\therefore z = 0$  OR

$z = \frac{(1 + 3i) \pm \sqrt{(1 + 3i)^2 + 16}}{2}$

$z = (1 + 3i) \pm \sqrt{8 + 6i}$

$z = 1 + 3i \pm \frac{3 + i}{2}$

$z = 0, 2 + 2i, -1 + i$  (3)

(c)  $z^4 = 2\sqrt{3} + 2i$

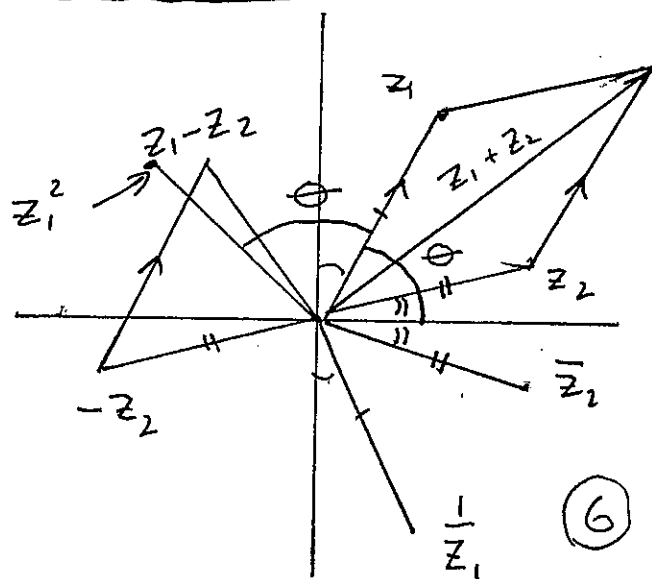
$z^4 = 16 \cos \pi/6$

$\therefore z = 16^{1/4} \cos \frac{2k\pi + \pi/6}{4}$   $k = -2, -1, 0, 1, 2$

$z = 2 \cos \frac{-23\pi}{24}, 2 \cos \frac{-11\pi}{24}, 2 \cos \frac{\pi}{24}$

$2 \cos \frac{13\pi}{24}$  (3)

Question 3.

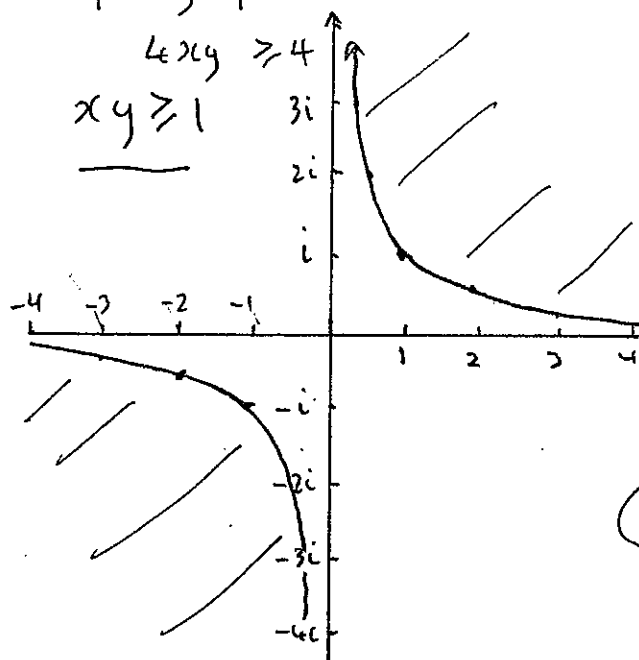


(6)

$$|4xyi| \geq 4$$

$$4xy \geq 4$$

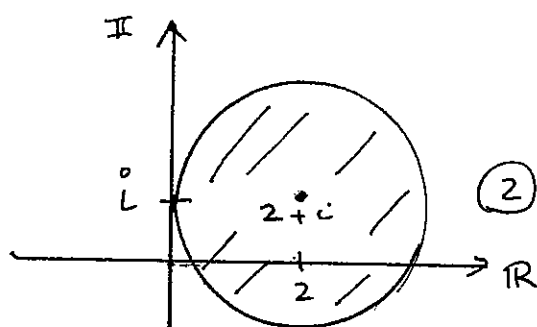
$$xy \geq 1$$



(3)

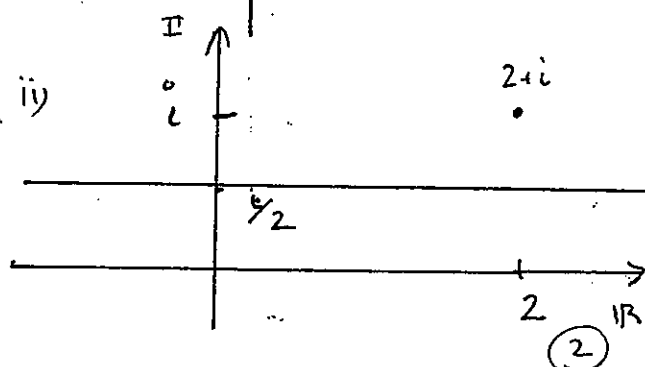
Question 4.

(a) (i)



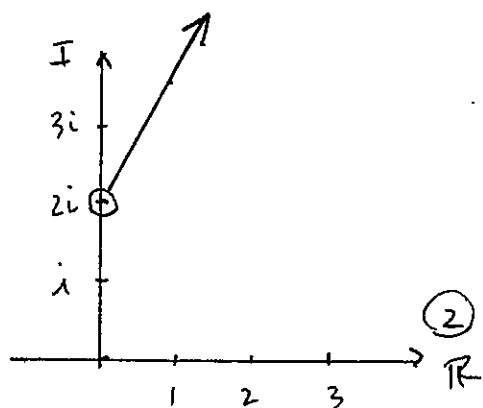
(2)

(ii)



(2)

(iii)



(2)

$$(iv) |z^2 - (\bar{z})^2| \geq 4$$

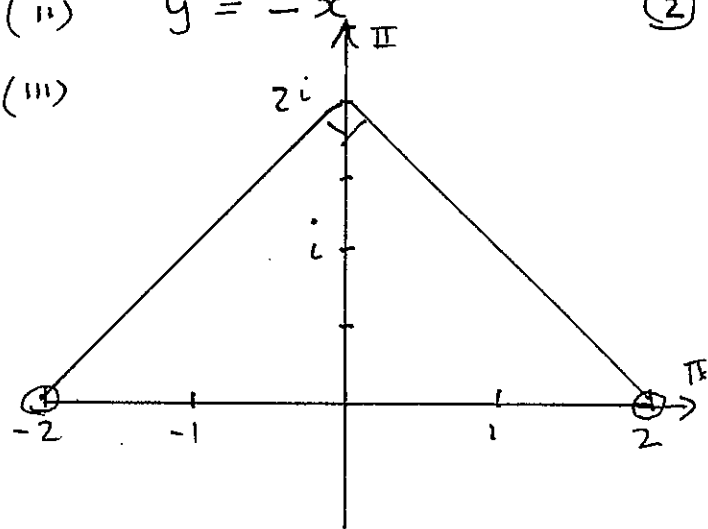
$$|(x+iy)^2 - (x-iy)^2| \geq 4$$

$$|x^2 - y^2 + 2xyi - (x^2 - y^2 - 2xyi)| \geq 4$$

$$(b) (i) (x-2)^2 + (y-1)^2 = 9. (2)$$

$$(ii) y = -x$$

(iii)



$$x^2 + (y-2)^2 = 8$$

$$y > 0$$

circle centre  $(0+2i)$

radius  $\sqrt{8}$

above real axis. (4)

(c)

$$\frac{1}{z} + \frac{1}{\bar{z}} = 1$$

$$\frac{1}{x+iy} + \frac{1}{x-iy} = 1$$

$$\frac{x-iy + x+iy}{x^2+y^2} = 1$$

$$\frac{2x}{x^2+y^2} = 1$$

$$2x = x^2 + y^2$$

$$0 = x^2 - 2x + y^2$$

$$1 = x^2 - 2x + 1 + y^2$$

$$(x-1)^2 + y^2 = 1$$

Circle centre  $(1,0)$   $\{1+0i\}$

Radius 1. (3)

### Question 5.

$$(a) z = \cos \theta + i \sin \theta$$

$$z^4 = \cos 4\theta + i \sin 4\theta \text{ (DE MOIVRE)}$$

$$z^4 = \cos^4 \theta + 4 \cos^3 \theta \sin \theta i - 6 \cos^2 \theta \sin^2 \theta - 4 \cos \theta \sin^3 \theta i + \sin^4 \theta$$

(BINOMIAL)

EQUATING REALS.

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta \quad (2)$$

EQUATING IMAGINARY

$$(b) \sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta \quad (2)$$

$$(c) \tan 4\theta = \frac{4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta}{\cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta}$$

$$(\div \cos^4 \theta)$$

$$= \frac{4 \tan \theta - 4 \tan^3 \theta}{1 - 6 \tan^2 \theta + \tan^4 \theta} \quad (2)$$

$$(d) z^n + \frac{1}{z^n} = 2 \cos n\theta$$

$$(\cos \theta + i \sin \theta)^n + \frac{1}{(\cos \theta + i \sin \theta)^n}$$

$$\cos n\theta + i \sin n\theta + \cos -n\theta + i \sin -n\theta$$

DEMOIVRE

But  $\cos$  is an even fn  
 $\sin$  is an odd fn

$$= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$$

$$= 2 \cos n\theta$$

(1)

Similarly

$$(e) \cos n\theta + i \sin n\theta - (\cos n\theta - i \sin n\theta)$$

$$= 2i \sin n\theta$$

(1)

$$(f) (2 \cos \theta)^6 = \left(z + \frac{1}{z}\right)^6$$

$$64 \cos^6 \theta = z^6 + 6z^4 + 15z^2 + 20 + \frac{15}{z^2} + \frac{6}{z^4} + \frac{1}{z^6}$$

(BINOMIAL)

$$64 \cos^6 \theta = z^6 + \frac{1}{z^6} + 6\left(z^4 + \frac{1}{z^4}\right) + 15\left(z^2 + \frac{1}{z^2}\right) + 20$$

$$64 \cos^6 \theta = 2 \cos 6\theta + 12 \cos 4\theta + 30 \cos 2\theta + 20$$

$$\cos^6 \theta = \frac{1}{32} \left\{ \cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 1 \right\} \quad (3)$$

### Question 6.

(a) (i)

$$\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \times \frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta + i \cos \theta}$$

$$= \frac{(1 + \sin \theta)^2 - \cos^2 \theta + 2(1 + \sin \theta)i \cos \theta}{(1 + \sin \theta)^2 + \cos^2 \theta}$$

$$= \frac{(1 + \sin \theta)^2 - (1 - \sin^2 \theta) + 2(1 + \sin \theta)i \cos \theta}{1 + 2 \sin \theta + \sin^2 \theta + \cos^2 \theta}$$

$$= \frac{(1 + \sin \theta)^2 - (1 + \sin \theta)(1 - \sin \theta) + 2(1 + \sin \theta) \cos \theta}{2(1 + \sin \theta)}$$

$$= \frac{(1 + \sin \theta) - (1 - \sin \theta) + 2 \cos \theta}{2}$$

$$= \frac{2 \sin \theta + 2 \cos \theta}{2} = \sin \theta + \cos \theta$$

(3)

4 Question to continue.

$$(1 + \sin \frac{\pi}{5} + i \cos \frac{\pi}{5})^5 + i(1 + \sin \frac{\pi}{5} - i \cos \frac{\pi}{5})^5 = 0$$

$$\div (1 + \sin \frac{\pi}{5} - i \cos \frac{\pi}{5})^5$$

$$\left( \frac{1 + \sin \frac{\pi}{5} + i \cos \frac{\pi}{5}}{1 + \sin \frac{\pi}{5} - i \cos \frac{\pi}{5}} \right)^5 + i = 0$$

$$\left( \sin \frac{\pi}{5} + i \cos \frac{\pi}{5} \right)^5 + i = 0$$

$$\left[ \cos \left( \frac{\pi}{2} - \frac{\pi}{5} \right) + i \sin \left( \frac{\pi}{2} - \frac{\pi}{5} \right) \right]^5 + i = 0$$

$$\left( \cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) + i = 0$$

DE MOIVRE

$$\cos -\frac{\pi}{2} + i \sin -\frac{\pi}{2} + i = 0$$

$$0 - i + i = 0 \quad (3)$$

b) (1)  $\omega^k = \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5}$

DE MOIVRE

$$z^5 - 1 = \left( \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5} \right)^5 - 1$$

$$= (\cos 2k\pi + i \sin 2k\pi) - 1$$

$$= 1 + 0i - 1$$

$$= 0 \quad (2)$$

$$z^5 - 1 = (z-1)(z^4 + z^3 + z^2 + z + 1)$$

IF  $\omega$  IS A COMPLEX ROOT

$$(\omega-1)(\omega^4 + \omega^3 + \omega^2 + \omega + 1) = 0$$

$$\text{But } \omega-1 \neq 0 \text{ (COMPLEX)}$$

$$\therefore \omega^4 + \omega^3 + \omega^2 + \omega + 1 = 0 \quad (2)$$

$$(iii) (\omega-1)(1+2\omega+3\omega^2+4\omega^3+5\omega^4)$$

$$= \omega + 2\omega^2 + 3\omega^3 + 4\omega^4 + 5\omega^5 - 1 - 2\omega - 3\omega^2 - 4\omega^3 - 5\omega^4$$

$$= -\omega - \omega^2 - \omega^3 - \omega^4 - 1 + 5\omega^5$$

0 + sin - 5 (5)

$$S = \frac{\omega}{1-\omega^2} + \frac{\omega^3}{1-\omega^4} + \frac{\omega^5}{1-\omega} + \frac{\omega^7}{1-\omega^2}$$

$$S = \frac{\omega}{1-\omega^2} \frac{\omega^3}{\omega^3} + \frac{\omega^2}{1-\omega^4} \frac{\omega}{\omega} + \frac{\omega^3}{1-\omega} + \frac{\omega^4}{1-\omega^2}$$

$$S = \frac{\omega^4}{\omega^3-1} + \frac{\omega^3}{\omega-1} + \frac{\omega^3}{1-\omega} + \frac{\omega^4}{1-\omega^2}$$

$$S = \frac{-\omega^4}{\omega^3-1} - \frac{\omega^3}{1-\omega} + \frac{\omega^3}{1-\omega} + \frac{\omega^4}{1-\omega^2}$$

$$S = 0. \quad (2)$$

Question 7

$$(a) \left( \frac{1+i \tan \theta}{1-i \tan \theta} \right)^n = \left[ \frac{\cos \theta + i \sin \theta}{\cos \theta} \cdot \frac{\cos \theta - i \sin \theta}{\cos \theta} \right]^n$$

$$= \left[ \frac{\cos \theta + i \sin \theta}{\cos \theta} \cdot \frac{\cos -\theta + i \sin -\theta}{\cos \theta} \right]^n = \frac{\cos n\theta + i \sin n\theta}{\cos -n\theta + i \sin -n\theta}$$

$$\left\{ \begin{array}{l} \cos \theta \text{ is an even fn} \\ \sin \theta \text{ is an odd fn} \end{array} \right\} = \frac{\cos n\theta + i \sin n\theta}{\cos n\theta - i \sin n\theta}$$

$$= \frac{\cos n\theta (1 + i \tan n\theta)}{\cos n\theta (1 - i \tan n\theta)}$$

$$= \frac{1 + i \tan n\theta}{1 - i \tan n\theta} \quad (3)$$

(b)  $z^6 - 1 = 0$

$$z^6 = 1$$

$$z^6 = \cos 2k\pi$$

$$z = \cos \frac{2k\pi}{6}$$

$$k = -3, -2, -1, 0, 1, 2, 3$$

$$z = \cos -\pi, \cos -\frac{2\pi}{3}, \cos -\frac{\pi}{3}, \cos 0, \cos \frac{\pi}{3}, \cos \frac{2\pi}{3} \quad (3)$$

$$z = -1, 1, \cos \frac{\pi}{3}, \cos -\frac{\pi}{3}, \cos \frac{2\pi}{3}, \cos -\frac{2\pi}{3}$$

$$(1) z^6 - 1 = (z-1)(z+1)(z - \cos \pi/3) \\ (z - \cos \pi/3)(z - \cos 2\pi/3)(z - \cos 4\pi/3)$$

for least arg  $\phi$  tangent to circle

Consider

$$(z - \cos \pi/3)(z - \cos 2\pi/3)$$

$$= [z - (\cos \pi/3 + i \sin \pi/3)][z - (\cos 2\pi/3 + i \sin 2\pi/3)]$$

$$= [z - (\cos \pi/3 + i \sin \pi/3)][z - (\cos \pi/3 - i \sin \pi/3)]$$

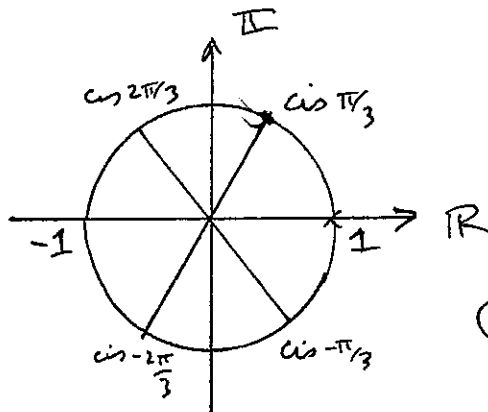
$$= z^2 - 2 \cos \pi/3 z + (\cos \pi/3 + i \sin \pi/3)(\cos \pi/3 - i \sin \pi/3)$$

$$= z^2 - 2 \cos \pi/3 z + \cos^2 \pi/3 + \sin^2 \pi/3$$

$$= (z^2 - 2 \cos \pi/3 z + 1)$$

$$\therefore z^6 - 1 = (z-1)(z+1) \quad (4)$$

$$(z^2 - 2 \cos \pi/3 z + 1)(z^2 - 2 \cos 2\pi/3 z + 1)$$



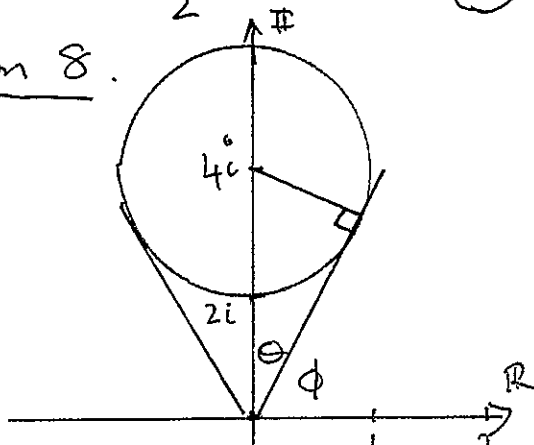
$$(iv) \text{ Area} = 6 \times \frac{1}{2} \times 1 \times 1 \times \sin 60$$

$$= 3 \times \frac{\sqrt{3}}{2}$$

$$= \frac{3\sqrt{3}}{2} u^2 \quad (2)$$

Question 8.

(a)



$$\therefore \sin \theta = \frac{3}{4}$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$\phi = 60^\circ (\pi/3) \quad (3)$$

$$\therefore \text{MAX ARG } (180 - 60 = 120) \quad (2\pi/3)$$

$$(b) \frac{2}{1+z} = \frac{2}{1 + \cos \theta + i \sin \theta}$$

$$= \frac{2}{1 + \cos \theta + i \sin \theta} \times \frac{1 + \cos \theta - i \sin \theta}{1 + \cos \theta - i \sin \theta}$$

$$= \frac{2(1 + \cos \theta - i \sin \theta)}{(1 + \cos \theta)^2 + \sin^2 \theta}$$

$$= \frac{z(1 + \cos \theta + i \sin \theta)}{z(1 + \cos \theta)}$$

$$= \frac{1 - i \sin \theta}{1 + \cos \theta}$$

$$= \frac{1 - i(\sin \theta/2 \cos \theta/2)}{1 + 2 \cos^2 \theta/2 - 1}$$

$$= 1 - i \tan \theta/2 \quad (3)$$

$$(c) \vec{OA} = 3 + i$$

$$\therefore \vec{OC} = -1 + 3i \quad (\perp OA)$$

$$\vec{OB} = \vec{OA} + \vec{OC}$$

$$= (1+i)(3+i)$$

$$= 2 + 4i \quad (3)$$

Question 9

$$|z_1 + z_2|^2 = (z_1 + z_2)(\overline{z_1 + z_2}) \quad (i)$$

$$= (z_1 + z_2)(\overline{z_1} + \overline{z_2})$$

$$= z_1 \overline{z_1} + z_1 \overline{z_2} + z_2 \overline{z_1} + z_2 \overline{z_2} \quad (i) \sin \theta + i \cos \theta = z$$

$$= |z_1|^2 + z_1 \overline{z_2} + z_2 \overline{z_1} + |z_2|^2$$

$$|z_1 - z_2|^2 = (z_1 - z_2)(\overline{z_1 - z_2})$$

$$= (z_1 - z_2)(\overline{z_1} - \overline{z_2})$$

$$= z_1 \overline{z_1} - z_1 \overline{z_2} - z_2 \overline{z_1} + z_2 \overline{z_2}$$

$$= |z_1|^2 - z_1 \overline{z_2} - z_2 \overline{z_1} + |z_2|^2$$

$$\therefore |z_1 + z_2|^2 + |z_1 - z_2|^2$$

$$= 2\{|z_1|^2 + |z_2|^2\} \quad (2)$$

$$(b) 1 + \cos \theta + i \sin \theta = z$$

$$(i) |z| = \sqrt{(1 + \cos \theta)^2 + (\sin \theta)^2}$$

$$= \sqrt{2 + 2 \cos \theta}$$

$$= \sqrt{2 + 2(2 \cos^2 \theta/2 - 1)}$$

$$(\cos 2A = 2 \cos^2 A - 1)$$

$$= \sqrt{4 \cos^2 \theta/2}$$

$$= 2 \cos \theta/2$$

$$\arg z = \tan^{-1} \frac{\sin \theta}{1 + \cos \theta}$$

$$= \frac{2 \sin \theta/2 \cos \theta/2}{1 + 2 \cos^2 \theta/2 - 1} \quad (3)$$

$$= \tan^{-1} (\tan \theta/2) = \frac{\theta}{2}$$

$$z = 2 \cos \frac{\theta}{2} (\cos \theta/2)$$

$$\therefore z^2 = 4 \cos^2 \theta/2 (\cos \theta)$$

$$= 4 \cos^2 \theta/2 (\cos \theta + i \sin \theta) \quad (1)$$

$$z = \cos(\pi/2 - \theta) + i \sin(\pi/2 - \theta)$$

$$|z| = 1 \quad \arg = (\pi/2 - \theta) \quad (2)$$

$$(ii) \sqrt{\sin \theta + i \cos \theta} = \pm \sin \frac{\pi/2 - \theta}{2}$$

$$= \pm \sin(\pi/4 - \theta/2) \quad (1)$$

Question 10

$$(i) z_1 = \cos \theta + i \sin \theta$$

$$z_2 = \cos \frac{2\pi}{3} (z_1)$$

$$z_3 = \cos -2\pi/3 (z_1) \quad (2)$$

ROTATION IN ARGAND

$$(ii) z_1 + z_2 + z_3 = (1 + \cos 2\pi/3 + \cos -2\pi/3) z_1$$

$$\text{Now } 1 + \cos 2\pi/3 + \cos -2\pi/3 = 1 + \omega + \omega^2$$

$\omega$  is cube root of unity

$$\therefore 1 + \omega + \omega^2 = 0 \quad (2)$$

$$\text{OR } 1 + \cos 2\pi/3 + i \sin 2\pi/3 + \cos -2\pi/3 + i \sin -2\pi/3$$

$$= 1 + \frac{1}{2} + \frac{\sqrt{3}}{2}i - \frac{1}{2} - \frac{\sqrt{3}}{2}i = 0$$

$$(iii) z_2 z_3 = \cos \frac{2\pi}{3} (z_1) \cdot \cos -\frac{2\pi}{3} (z_1)$$

$$= z_1^2$$

$$\text{similarly } z_1 z_3 = z_2^2 \quad z_1 z_2 = z_3^2$$

$$\therefore z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_1 z_3 + z_2 z_3$$

(3)